

args)
1");



```
Public Static main ( ) {  
    _____  
    _____  
    _____  
    return _____;  
}
```

```
System.out.print("A");  
System.out.println("B");  
System.out.println("C");
```



Methods

```
public class  
public static void main(String[] args)
```

```
draw()  
System.out.println("Java");  
draw();
```

```
Public static void draw()  
{  
for (int i=1; i<=5; i++)  
{  
System.out.print("*");  
}}
```

①

```
*****  
Java  
*****
```

②

```
*****  
Java  
#####
```

Char X = '*' ;
#

```
*****  
Java  
*****
```

Methods

```
public class  
{  
    public static void main(String[] args)  
    {  
        draw('X');  
        System.out.println("In java");  
        draw('X');  
    }  
}
```

```
public static void draw(char X)  
{  
    for (int i=1; i<=5; i++)  
    {  
        System.out.print(X);  
    }  
}
```

①

```
*****  
Java  
*****
```

②

```
*****  
Java  
#####
```

```
***  
Java  
#####
```

```
*****  
Java  
*****
```

```
*****  
Java  
#####
```


Methods

Write a program using three methods
sum of 3 numbers, average, display

```
import java.util.Scanner;
public class           
{
    public static void main (String[] args)
    {
        Scanner input = new Scanner (System.in);
        sy            ("Enter the three numbers");
        int n1 = input.nextInt();
        int n2 = input.nextInt();
        int n3 = input.nextInt();
        int t = sum(n1, n2, n3);
        double avg = average(n1, n2, n3);
        display (t, avg);
    }
}
```

```
public static int sum (int x, int y, int z)
{
    return x + y + z;
}
```

```
public static double average (int a, int b, int c)
{
    return sum(a, b, c) / 3.0;
}
```

```
public static void display (int s, double v)
{
    System.out.println ("Sum = " + s);
    System.out.println ("average = " + v);
}
```

5/2

Return

رجع الى الـ Call

(Void)

رجع الى الـ Call
الى الـ Call

Methods

to a program using three methods
sum of 3 numbers, average, display

Enter the three numbers
5
3
4
Sum=12
average=4.0

```
import java.util.Scanner;
public class ...
{
    public static void main (String [] args)
    {
        Scanner input = new Scanner (System.in);
        // Enter the three numbers
        int n1 = input.nextInt();
        int n2 = input.nextInt();
        int n3 = input.nextInt();
        int t = sum(n1, n2, n3);
        double avg = average(t, n1, n2, n3);
        display(t, avg);
    }

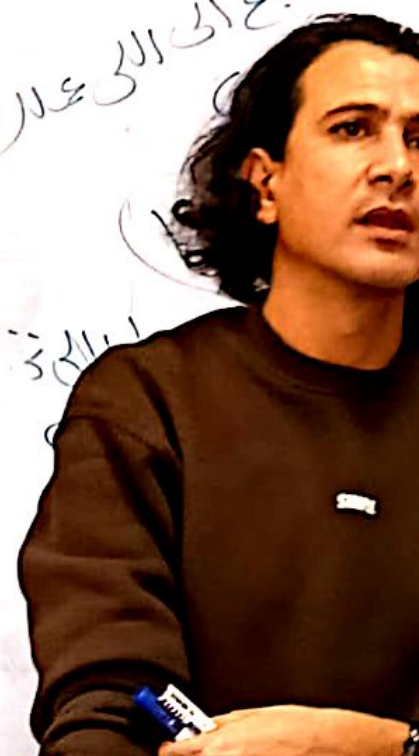
    public static int sum (int x, int y, int z)
    {
        return x + y + z;
    }

    public static double average (int a, int b, int c)
    {
        return (sum(a, b, c) / 3.0);
    }

    public static void display (int s, double v)
    {
        System.out.println("Sum=" + s);
        System.out.println("average=" + v);
    }
}
```

5/2

return



Methods

Public static void main (String[] args)

```
{
    int x=3;
    int y = m1(x);
    Sy (y);
}
```

Public static int m1(int n)

```
{
    n++;
    int z = m2(n);
}
```

Return z;

Public static int m2(int m)

```
{
    m *= 2;
    return m;
}
```

Scope of variables

local variable
global variable

main

```
{
    int x=3;
    display();
}
```

display ()

```
{
    Sy (x);
}
```

main

```
{
    int x=3;
    display(x);
}
```

main

```
{
    int x=5;
    {
        int y=3;
    }
}
```

System.out.println(y);

main

```
{
    int i;
    for (i=1; i=3; i++)
    {
        Sy ("Hello");
    }
    Sy (i);
}
```

Call by value

main

```
{
    int x=5;
    int y=3;
    Sy ("X="+x);
    Sy ("y="+y);
    display(x, y);
}
```

display (int y, int x)

```
{
    Sy ("X="+x);
    Sy ("y="+y);
}
```

X=5
y=3
X=3
y=5

ables
table
variable

double x = display(3, 5)

3;
);

~~int
double~~

display(int a, int b)
display(int a, int b)

(X);
?

?

int

Scope of variables

- local variable
- global variable

double x = display(3);

Ex

```
{  
  int x=3;  
  display();  
}
```

3)

```
display();
```

3)

```
{  
  Sy — (x);  
}
```

```
int x=3;  
display(x);
```

```
display(int x);  
Sy — (x);
```

double

display(int x, int y)

display(int a, int b)

ain
it i,

```
for(i=1; i<=3; i++)  
{  
  Sy — ("Hello");  
}
```